

Technology Stack

Date	13 July 2026
Team ID	N/A
Project Name	FinRelief AI – AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs used to build the solution. A well-chosen technology stack ensures that the application is scalable, maintainable, and aligned with the project requirements.

The table below details the specific technologies chosen for each component of the FinRelief AI architecture and the justification for each choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js	Component-based architecture allows a responsive, reusable dashboard and loan-entry UI, with fast rendering for financial visuals.
2	Backend / Server-Side	Python (FastAPI)	High-performance async framework with built-in validation, ideal for exposing the financial engine and AI negotiation services as REST APIs.
3	Database / Data Storage	SQLite with SQLAlchemy ORM	Lightweight, zero-config relational storage that is easy to set up for the project scale, with ORM support for clean, maintainable data models.
4	Cloud / Hosting / Deployment	To be deployed on a cloud platform (e.g., Render/Heroku/AWS)	Enables easy scalability and public accessibility of the borrower-facing web application once finalized.
5	Version Control & CI/CD	GitHub	Enables seamless team collaboration, code review, and version history tracking across the team.
6	Third-Party APIs / Other Tools	Google Gemini API	Powers the AI-driven negotiation letter generation and settlement recommendation features based on borrower financial data.